DAY 6 | PRE-READ

# Connections, Promises and Manuals

A light read before Day 6. About 15 minutes, no code required.

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## Why read this?

On Day 5 your API kept data in memory, so it vanished on restart. Today you connect it to a real database, link related data, make sure changes are safe, and write documentation that others enjoy reading.

## 1. A translator between two languages

Python thinks in objects. Most databases think in tables. An **ORM** (Object Relational Mapper) is a translator between them. You work with Python objects, and the ORM writes the database commands for you. We use one called **SQLAlchemy**.

| Python side | Database side |
|---|---|
| Class | Table |
| Attribute | Column |
| Object | Row |

## 2. How things relate

| Relationship | Everyday example |
|---|---|
| One-to-one | One person has one passport. |
| One-to-many | One teacher teaches many classes; each class has one teacher. |
| Many-to-many | Students join many clubs, and each club has many students. |

For many-to-many, you need a register in the middle: a sign-up sheet that lists which student joined which club. Databases use exactly this idea, called a link table.

## 3. All or nothing

When you transfer money, two things must happen: your account goes down and your friend's goes up. If only the first step happened, money would vanish. Databases protect you with **transactions**: a group of steps that either all succeed or are all undone.

> **Key idea:** commit means "make it permanent". Rollback means "undo everything since we started".

## 4. A manual that is always up to date

FastAPI builds its documentation directly from your code, so it is always current. The standard behind it is called **OpenAPI**, and the interactive page at /docs is called **Swagger UI**. Good documentation answers three questions quickly: what can I do, what must I send, and what will I get back?

## 5. Think about these before class

- List three relationships in a telecom support desk (customers, tickets, agents) and classify each.
- Describe a situation in your job where a half-finished change would cause problems.
- What makes a manual or guide easy to follow?

Tsunagari (繋がり): connection, the links between things